

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: METALLURGICAL AND MATERIALS ENGINEERING

2. Subject Name: XRD and Microscopy Lab
Credits:2

L-T-P: 0-0-3

3. Pre-requisites: No

4. Syllabus and reference books:

Syllabus:

Module/ Experiment I

Production and Measurement of X-Rays and Indexing of Powder Patterns of Cubic Metals.

Aims:

Understanding constructional features of an x-ray machine and production of x-ray and indexing.

Module/ Experiment II

Estimation of Crystal Size and R.M.S. Strain in Mechanically Alloyed Nanostructured Metals.

Aims: To determine the crystal size and strain of ball milled Al powders.

Module/ Experiment III

Determination of Retained Austenite in Quenched Steel.

Module/Experiment IV:

Stereographic Projection

Aims: To understand and use of stereographic projection for the representation of angular relationships.

Module/ Experiment V

Optical Microscopy and Quantitative Analysis

Module VI/Experiment VI

Scanning Electron Microscopy and Elemental Analysis

Aims: To learn about principal of operation of Scanning electron microscope and elemental analysis.

Inculcate practical expertise on use and interpretation of Scanning electron micrograph.

Module VII/ Experiment VII

Atomic Force Microscopy

Module VIII/ Experiment VIII

Transmission Electron Microscopy

Reference Books:

1. B.D. Cullity, Elements of X-ray Diffraction, 2nd Addison-Wesley, 1978
2. Metallography: An Introduction to the Study of the Structure of Metals, Chiefly by the Aid of the

Microscope, by Arther Horseman Hiorns
 3. Metallographic Laboratory Practice, George F. Vander Voort, McGraw-Hill U.S.A
 4. Metallographic Laboratory Practice, George L. Kehl, 3rd edition, McGraw-Hill, U.S.A
 5. Transmission electron microscopy of materials, G. Thomas, M.J.Goringe, John, Wiley & Sons, 1979.
 6. Physical Principles of Electron Microscopy, R.F. Egerton, 2nd Edition, Springer.
 7. Compact Introduction to Electron Microscopy, Goerg H. Michler, Springer
 8. Scanning Electron Microscopy and X-Ray Microanalysis, Joseph I. Goldstein, Dale E. Newbury, Joseph R. Michael, Nicholas W.M. Ritchie, John Henry J. Scott, David C. Joy, 4th Edition Springer.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures/ lab hrs
1.	Production and Measurement of X-Rays and Indexing of Powder Patterns of Cubic Metals.	6
2.	Estimation of Crystal Size and R.M.S. Strain in Mechanically Alloyed Nanostructured Metals.	3
3.	Determination of Retained Austenite in Quenched Steel.	3
4.	Stereographic Projection	6
5	Optical Microscopy and Quantitative Analysis	3
6	Scanning Electron Microscopy and Elemental Analysis	3
7	Atomic Force Microscopy	3
8	Transmission Electron Microscopy	6
	Viva/test	6
Total number of hours		36